

Fictitious Play

1. It is commonly supposed that adaptive behavior will lead intelligent people to eventually coordinate on mutually consistent outcomes.
 - a. This view is often alluded to by economists who are about to base their analysis of some strategic situation on the logic of strategic rationality.
 - b. Economists who bother with this motivation assume that a more complete analysis would explain the origin of mutually consistent behavior;
 - c. And, hence, it is reasonable to use the logic of strategic rationality as a short cut for finding equilibria that would result from the more complicated analysis.
 - d. The most careful statement of this view that I know of is Lucas's paper, "Adaptive Behavior and Economic Theory," in *Rational Choice*, eds. R.M. Hogarth and M.W. Reder, (Chicago: University of Chicago Press, 1987)
2. However, it turns out to be surprisingly difficult to specify an explicit dynamical system that always converges to an equilibrium determined by the strategic analysis.
 - a. When behavior converges to a limit cycle, the adaptive behavior being modeled is perfectly predictable and easily exploitable.
 - b. It is often possible to stabilize a dynamic by using more history than the current state.



3. Fictitious Play.

- a. Assume that players best respond to the empirical frequency distribution of their opponents past play.
- b. Robinson (1951) proved that the empirical frequency distribution of past play converges to an equilibrium distribution for any two-player, finite-strategy, zero-sum game.
- c. Recently, fictitious play has undergone something of a renaissance.

4. Two-player fictitious Play

- a. Stage game: $I = \{1,2\}$; Finite strategy spaces S^1, S^2 ; Payoff functions u^1, u^2
- b. Myopic maximizing assumption: Players choose their actions in each period to maximize that periods expected payoff given beliefs based on the empirical frequency of past play and an initial prior.

c. *Assessment*

- i. Player i has an exogenous initial weight function $\kappa_0^i: S^{-i} \rightarrow \mathbb{R}_+$
- ii. The weights are updated by adding 1 to the weight of each opponent strategy each time it is played

$$\kappa_t^i(s^{-i}) = \kappa_{t-1}^i(s^{-i}) + \begin{cases} 1 & \text{if } s_{t-1}^{-i} = s^{-i}, \\ 0 & \text{if } s_{t-1}^{-i} \neq s^{-i} \end{cases}$$

- iii. The probability that player i assigns to player $-i$ playing s^{-i} at date t is given by

$$\gamma_t^i(s^{-i}) = \frac{\kappa_t^i(s^{-i})}{\sum_{\tilde{s}^{-i} \in S^{-i}} \kappa_t^i(\tilde{s}^{-i})}$$

- iv. *Fictitious play* is any rule p that assigns $p_t^i(\gamma_t^i) \in BR^i(\gamma_t^i)$, where BR denotes best response.

5. Rationalizing Fictitious Play

- a. Robinson originally developed fictitious play as a way to solve games.
 - i. Fictitious refers to the idea that the play takes place in one's head, before the player selects a strategy in the stage game.
 - ii. If the algorithm converges to a Nash equilibrium of the stage game, then the deductive libration leads to a prescription for playing the stage game.
- b. Fudenberg & Levine develop the interpretation that Fictitious Play is a form of myopic Bayesian rationality,
 - i. Players assume the distribution of actions they are playing against is a sequence of i.i.d. multinomial random variables with a fixed but unknown distribution
 - ii. These beliefs are “rational” only if their opponents have converged to an equilibrium.
 - iii. If play converges, then the stationarity assumption required to rationalize fictitious play can be defended as being correct asymptotically.

6. Asymptotic Behavior of Fictitious Play

Proposition 2.1:

- (1) If s is a strict Nash equilibrium, and s is played at date t in the process of fictitious play, then s is played at all subsequent dates, that is, strict Nash equilibria are absorbing for the fictitious play dynamic.
- (2) Any pure-strategy steady state of fictitious play must be a Nash equilibrium.

Proof:

- [1] First, suppose that players' assessments γ_t^i are such that their optimal choices correspond to a strict Nash equilibrium σ .
- [2] Then, when the strict equilibrium is played, each player i 's beliefs in period $t+1$ are a convex combination of γ_t^i and a point mass on σ^{-i} :

$$\gamma_{t+1}^i = (1-\alpha_i)\gamma_t^i + \alpha_i\delta(\sigma^{-i})$$

- [3] Since expected utilities are linear in probabilities,

$$u^i(\sigma^i, \gamma_{t+1}^i) = \alpha_i u^i(\sigma^i, \delta(\sigma^{-i})) + (1-\alpha_i)u^i(\sigma^i, \gamma_t^i)$$

- [4] So if σ^i is a strict best response to γ_t^i for player i , then it is a strict best response to γ_{t+1}^i .

Conversely,

- [5] if play remains at a pure-strategy combination, then eventually the assessments will become concentrated at that combination, and so if the combination is not a Nash equilibrium, one of the players would eventually want to deviate, which is a contradiction.

7. In games without a pure strategy equilibrium, the fictitious play dynamic can not converge to any pure strategy combination.
8. Nevertheless, the empirical distribution over player i 's strategies can converge to a mixed equilibrium. Let the empirical distribution of j 's play be denoted

$$d_t^j(s^j) = \frac{\kappa_t(s^j) - \kappa_0(s^j)}{t}$$

Proposition 2.2: Under fictitious play, if the empirical distribution d_t^i over each player's choices converge, then the strategy combination corresponding to the product of these distributions is a Nash equilibrium.

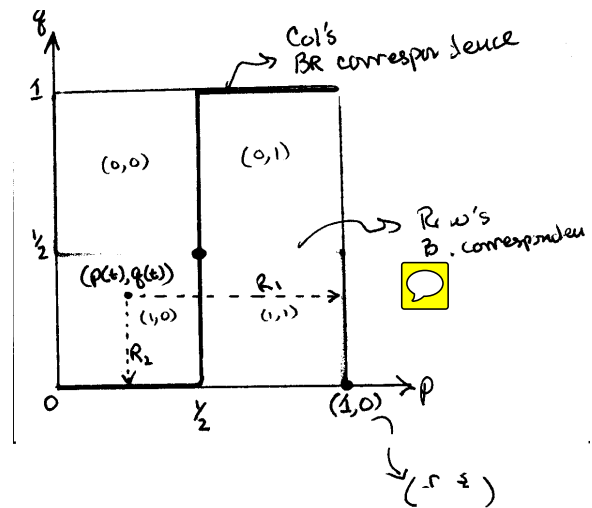
Proposition 2.3: Under fictitious play the empirical distributions converge if the stage game has generic payoffs and is 2×2 , or zero sum, or is solvable by iterated strict dominance, or has strategic complements and satisfies an other technical condition.

9. Example of Fictitious Play.

- a. Consider the following 2x2 game:

	s1	s2	
r1	0,2	3,0	(1-p)
r2	2,1	1,3	p
	(1-q)	q	

- b. The best response correspondences are given in figure 2.

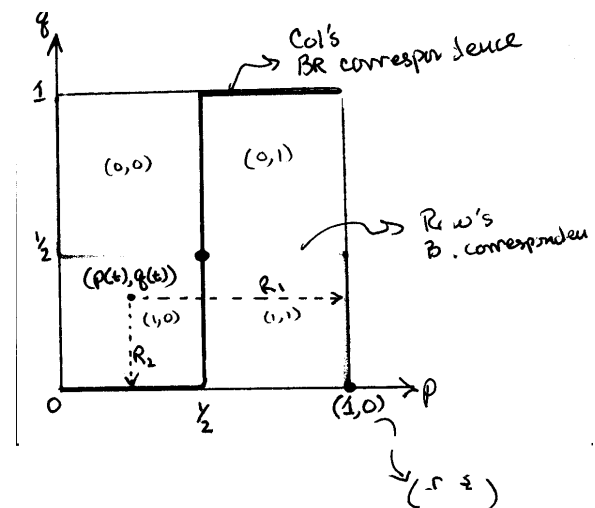


- c. The unique Nash equilibrium of the game is $\{1/2, 1/2\}$.

10. Fictitious Play assumptions:

- The game is played repeatedly by myopic players, that is, by players that don't use repeated game strategies. Instead,
 - Each player estimates the probability that his opponent will use either of their pure strategies.
 - The players choose their own pure strategy for the next stage so as to maximize their expected stage game payoff.
 - This calculation is equivalent to finding a best response to a mixed strategy in the stage game.
- How does the row player estimate the probability the column player will

- use her pure strategy s_2 , denoted q^e ?
- i. Suppose that t stages have been played;
 - ii. And that she has used s_2 m times;
 - iii. Then, the frequency with which she has used s_2 is m/t , denoted $q(t)$;
 - iv. The fictitious play assumption is that $q^e = q(t)$.
- c. Similarly, the column player sets $p^e = p(t)$, where $p(t)$ is the frequency the row player has played r_2 .
 - d. For the $(p(t), q(t))$ shown, the best response to the empirical frequency $(p(t), q(t))$ is (r_2, s_1) , see figure.
 - e. Notice that if the player's know how their opponent is forming beliefs they can do better by best responding to their opponent's best response to their own historical frequency of play.
 - f. That is, behavior is perfectly predictable and easily exploitable.



11. Switch to continuous time.

a. Let $(p(t), q(t))$ denote the frequency of (r_2, s_2) up to time t .

b. Divide the phase space into four regions according to the best response correspondences, see figure 3:

- i. $A \rightarrow (r_2, s_1) = (1, 0)$,
- ii. $B \rightarrow (r_2, s_2) = (1, 1)$,
- iii. $C \rightarrow (r_1, s_2) = (0, 1)$,
- iv. $D \rightarrow (r_1, s_2) = (0, 0)$.

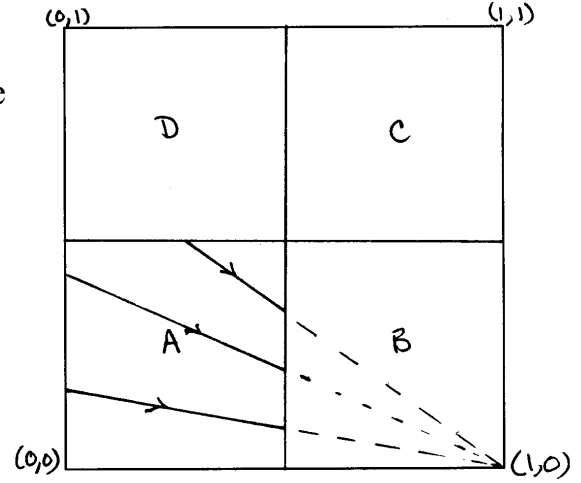


Figure 4: Best Response Regions.

c. If $(p(t), q(t)) \in A$, then it will continue to lie inside A until time $t+\tau$.

- i. Thus, the row player will be using r_2 from t until $t+\tau$,
- ii. And the column player will be using s_1 from t until $t+\tau$.

12. How many times will the row player have used r_2 at time $t+\tau$.

- a. Let λT denote the number of games played during a period of length T .
- b. Then he will have played r_2 $\lambda t p(t)$ times up to time t .
- c. Between t and $t+\tau$ he plays $\lambda \tau$ games, using r_2 on each occasion.
- d. Thus, r_2 is played $\lambda t p(t) + \lambda \tau$ times between 0 and $t+\tau$.
- e. The frequency of r_2 is thus

$$p(t+\tau) = \frac{\lambda t p(t) + \lambda \tau}{\lambda(t+\tau)} = \frac{t p(t) + \tau}{t+\tau}$$

- f. Now we want to know how $p(t)$ is changing with time and it is convenient to use the time derivative of $p(t)$, which is denoted $p'(t)$ and is defined

$$p'(t) = \lim_{\tau \rightarrow 0} \frac{p(t+\tau) - p(t)}{\tau}$$

- g. Now substitute the frequency of r_2 into the definition and solve for the differential equation:

$$\begin{aligned} p'(t) &= \lim_{\tau \rightarrow 0} \frac{p(t+\tau) - p(t)}{\tau} \\ &= \lim_{\tau \rightarrow 0} \frac{1}{\tau} \left(\frac{tp(t) + \tau}{t+\tau} - p(t) \right) \\ &= \lim_{\tau \rightarrow 0} \left(\frac{tp(t) + \tau - tp(t) - \tau p(t)}{\tau(t+\tau)} \right) \\ &= \lim_{\tau \rightarrow 0} \frac{1 - p(t)}{t+\tau} \\ &= \frac{1 - p(t)}{t} \end{aligned}$$

- h. Now derive the differential equation for the column player: recall that she only uses s_1 between t and $t+\tau$: Hence, the frequency of s_2 between 0 and $t+\tau$ is

$$q(t+\tau) = \frac{\lambda t q(t) + \lambda \tau 0}{\lambda(t+\tau)} = \frac{t q(t)}{t+\tau}.$$

- i. Hence,

$$\begin{aligned} q'(t) &= \lim_{\tau \rightarrow 0} \frac{q(t+\tau) - q(t)}{\tau} \\ &= \lim_{\tau \rightarrow 0} \frac{1}{\tau} \left(\frac{t q(t)}{t+\tau} - q(t) \right) \\ &= \lim_{\tau \rightarrow 0} \left(\frac{t q(t) - t q(t) - \tau q(t)}{\tau(t+\tau)} \right) \\ &= \frac{-q(t)}{t} \end{aligned}$$

13. Solving the Differential Equations:

$$p'(t) = \frac{1 - p(t)}{t}$$

$$q'(t) = \frac{-q(t)}{t}$$

- a. Notice that we can write these equations as

$$tp'(t) + p(t) = 1$$

$$tq'(t) + q(t) = 0$$

- b. Notice that this can be written as

$$\frac{d}{dt}(tp(t)) = 1$$

$$\frac{d}{dt}(tq(t)) = 0$$

- c. Integrating both sides gives

$$tp(t) = t - a \quad \Rightarrow \quad 1 - p(t) = \frac{a}{t}$$

$$tq(t) = b \quad \Rightarrow \quad q(t) = \frac{b}{t}$$

- d. Dividing the two equations gives

$$\frac{1 - p(t)}{q(t)} = \frac{a}{b} \quad \text{or} \quad q(t) = \frac{b}{a}(1 - p(t))$$

- e. The points $(p(t), q(t))$ that satisfy this equation lie on a straight line that passes through the point $(1,0)$ in figure 3.
- f. Points in region A all move along straight line trajectories toward the target point $(1,0)$.

14. When a trajectory reaches the boundary a new system of differential

equations needs to be solved.

- a. In region B, the target point becomes $(1,1)$;
- b. In region C, the target point becomes $(0,1)$;
- c. In region D, the target point becomes $(0,0)$.

15. The important observation is that the process converges to $\{\frac{1}{2}, \frac{1}{2}\}$.

- a. Thus, $\{\frac{1}{2}, \frac{1}{2}\}$ is a global attractor.
- b. And it is the Nash equilibrium of the game.

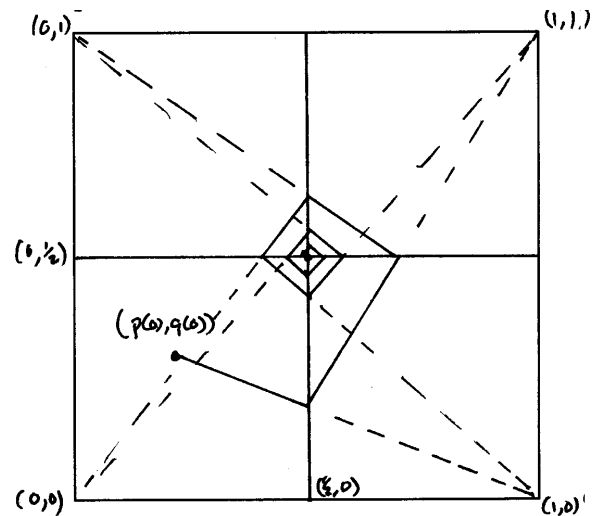


Figure 5: Convergent path

- c. However, notice that it is a Nash equilibrium in beliefs not in strategies.
- d. The players' are using pure strategies by assumption.

16. Shapley's Counter-Example:

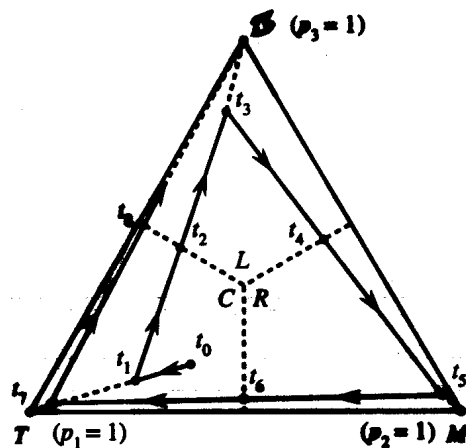
- a. Robinson's convergence result does not hold for non-zero sum games with more than two strategies.

	L	C	R
T	0,0	0,1	1,0
M	1,0	0,0	0,1
B	0,1	1,0	0,0

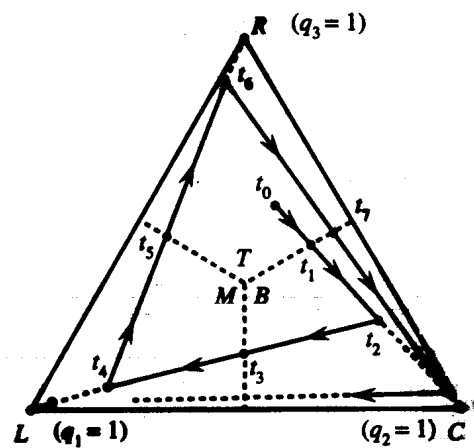
- b. Let (p_1, p_2, p_3) be the probability of T, M, and B for the row player.
- c. Let (q_1, q_2, q_3) be the probability of L, C, and R for the column player.
- d. Each player's mixed strategy is then a point on the 3-simplex, S^3 .
- e. The unique Nash equilibrium is $\{(1/3, 1/3, 1/3), (1/3, 1/3, 1/3)\}$.

17. Fictitious Play:

- a. Now let the vector $p(t)$ represent the vector of the frequency of the three row actions.
- b. And let the vector $q(t)$ represent the vector of the frequency of the three column actions.
- c. For this example we need to use two simplexes to keep track of how the frequencies are moving.



Row Player's Freq.



Column Player Freq.

